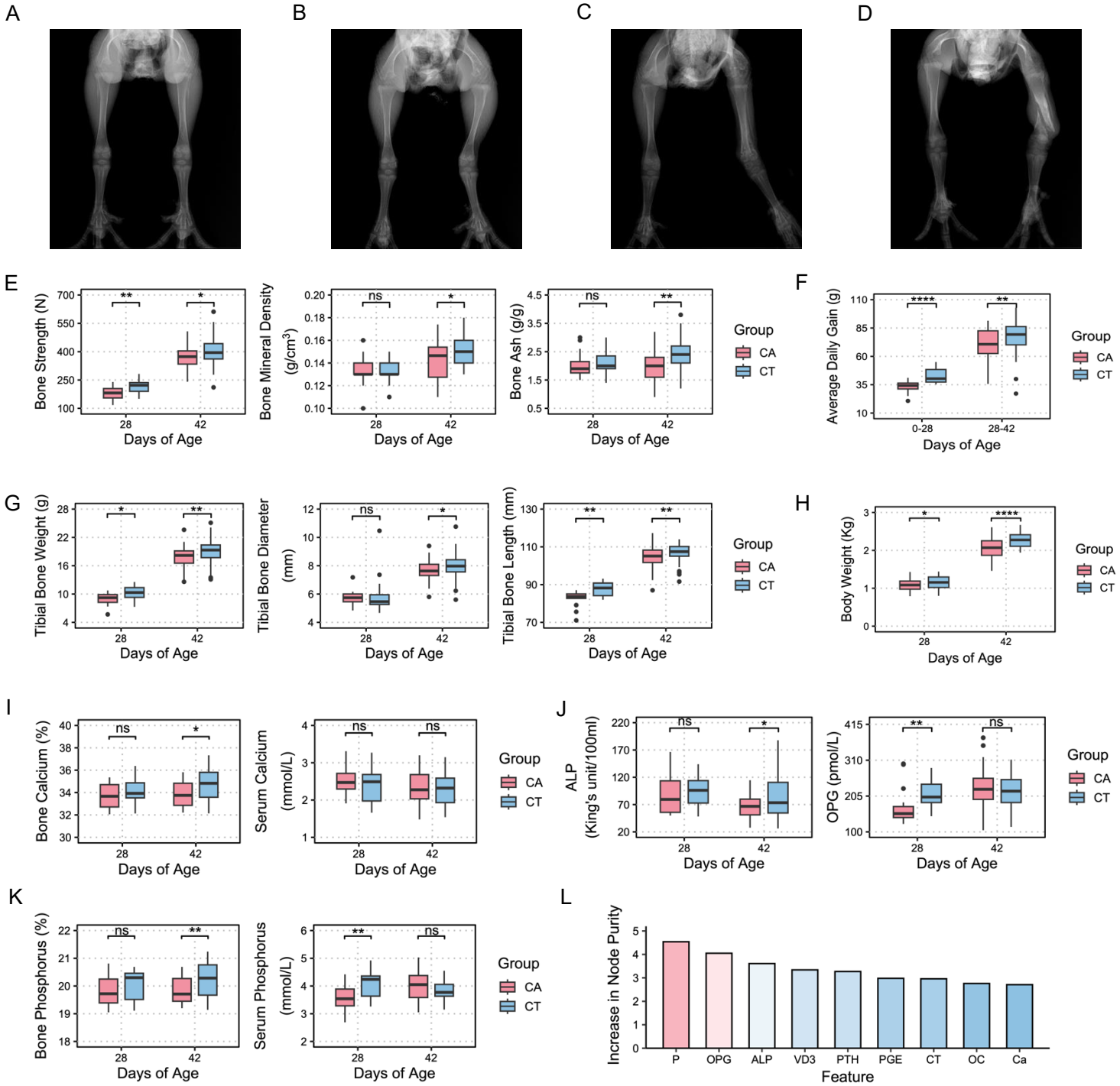
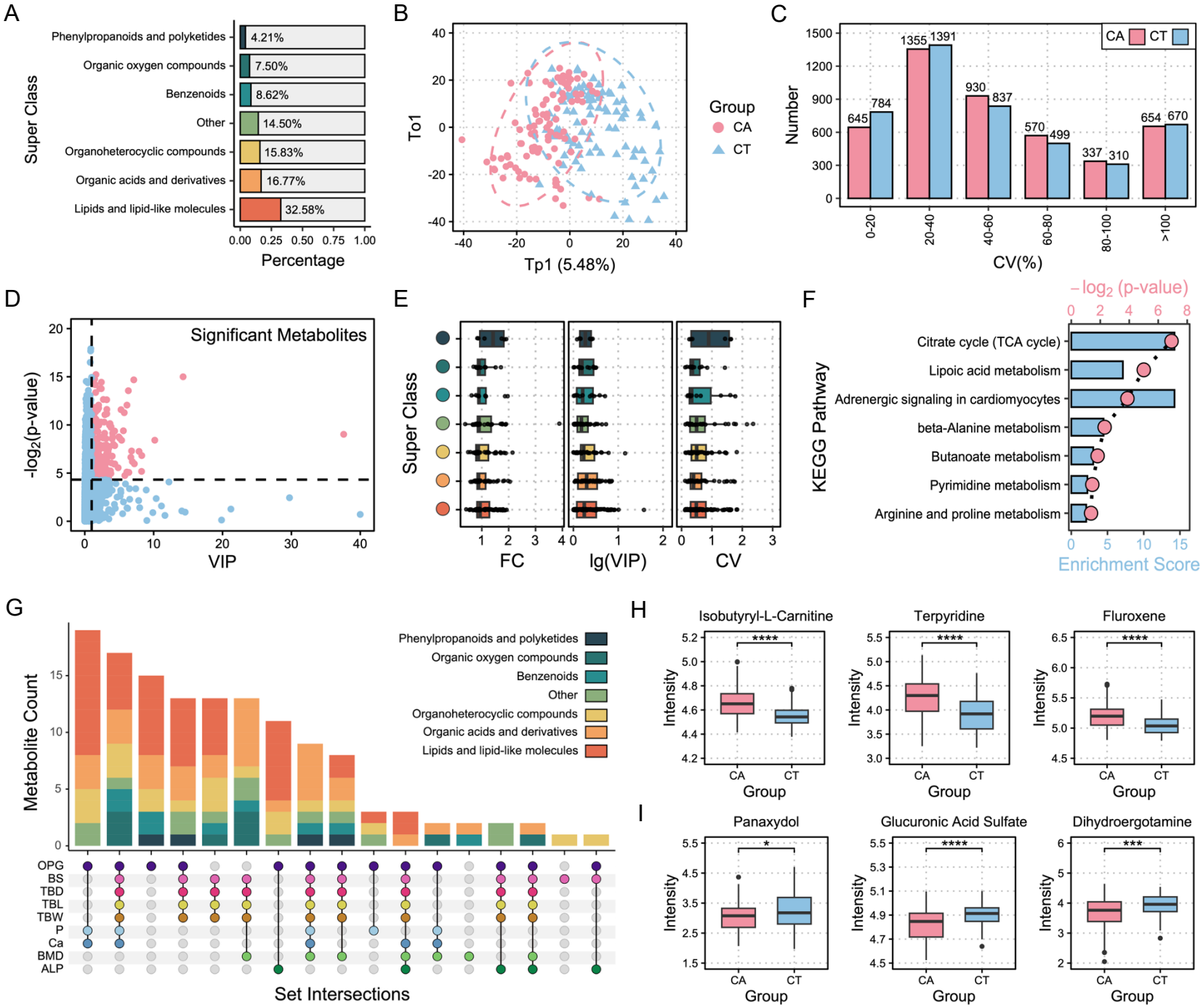
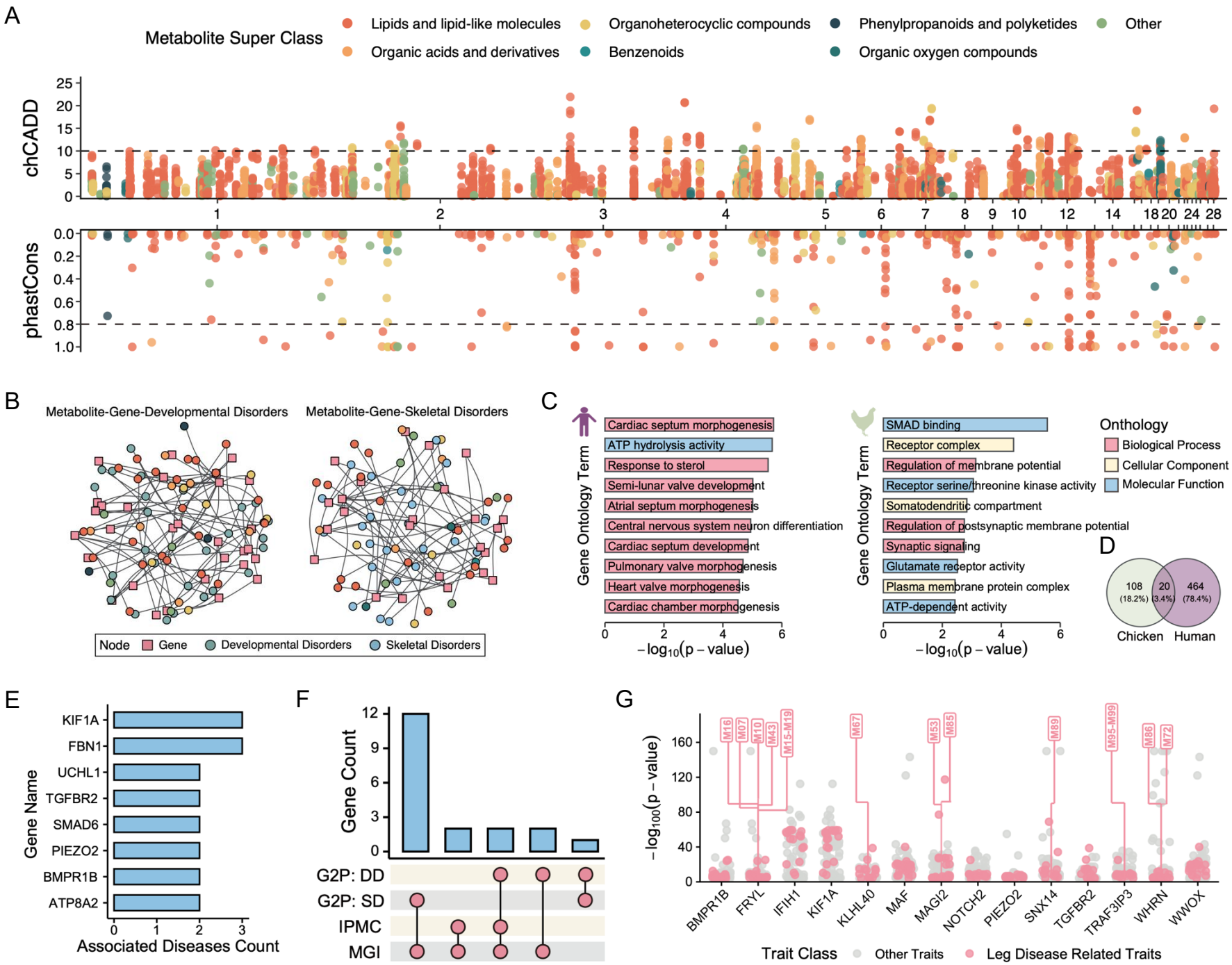
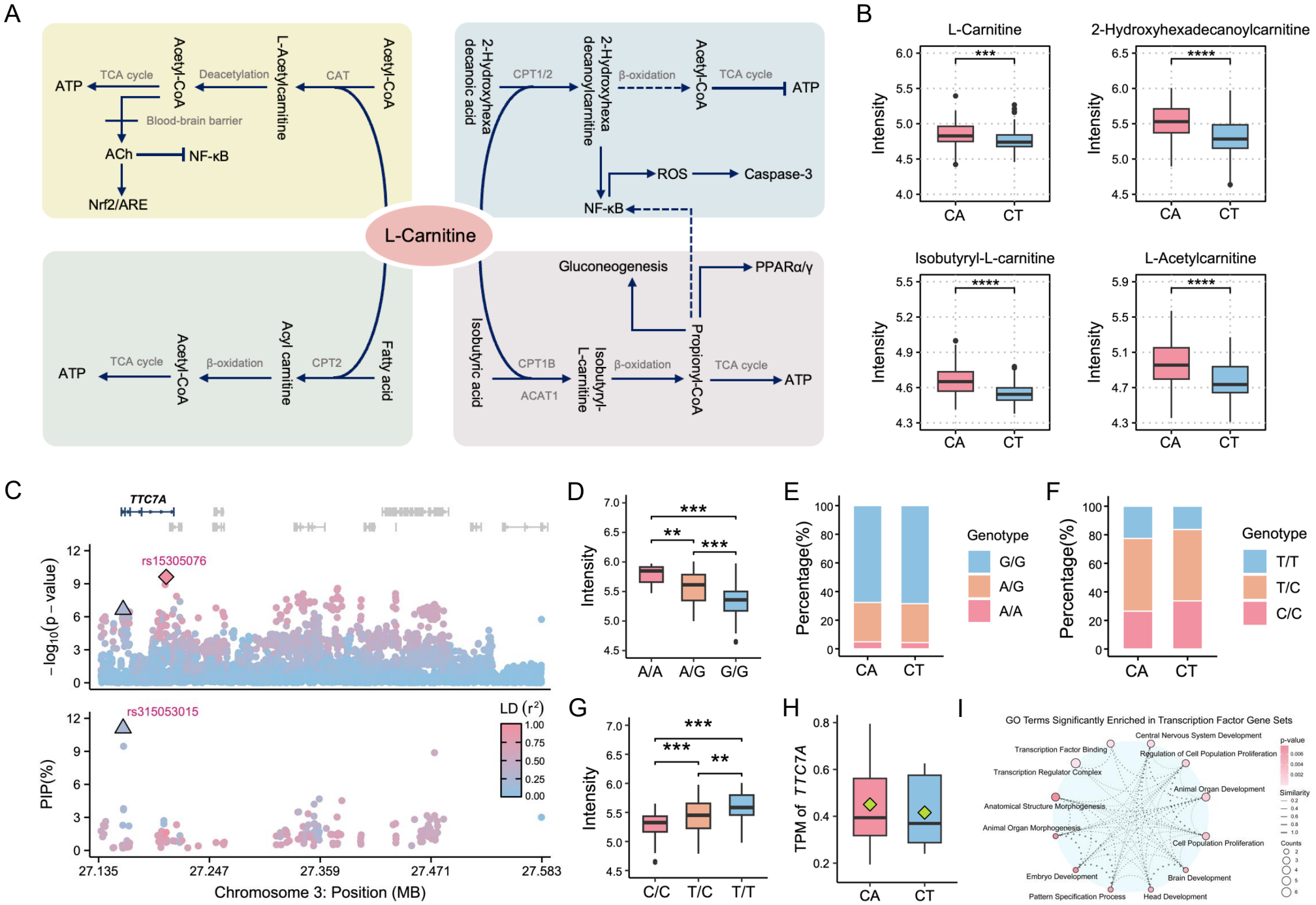
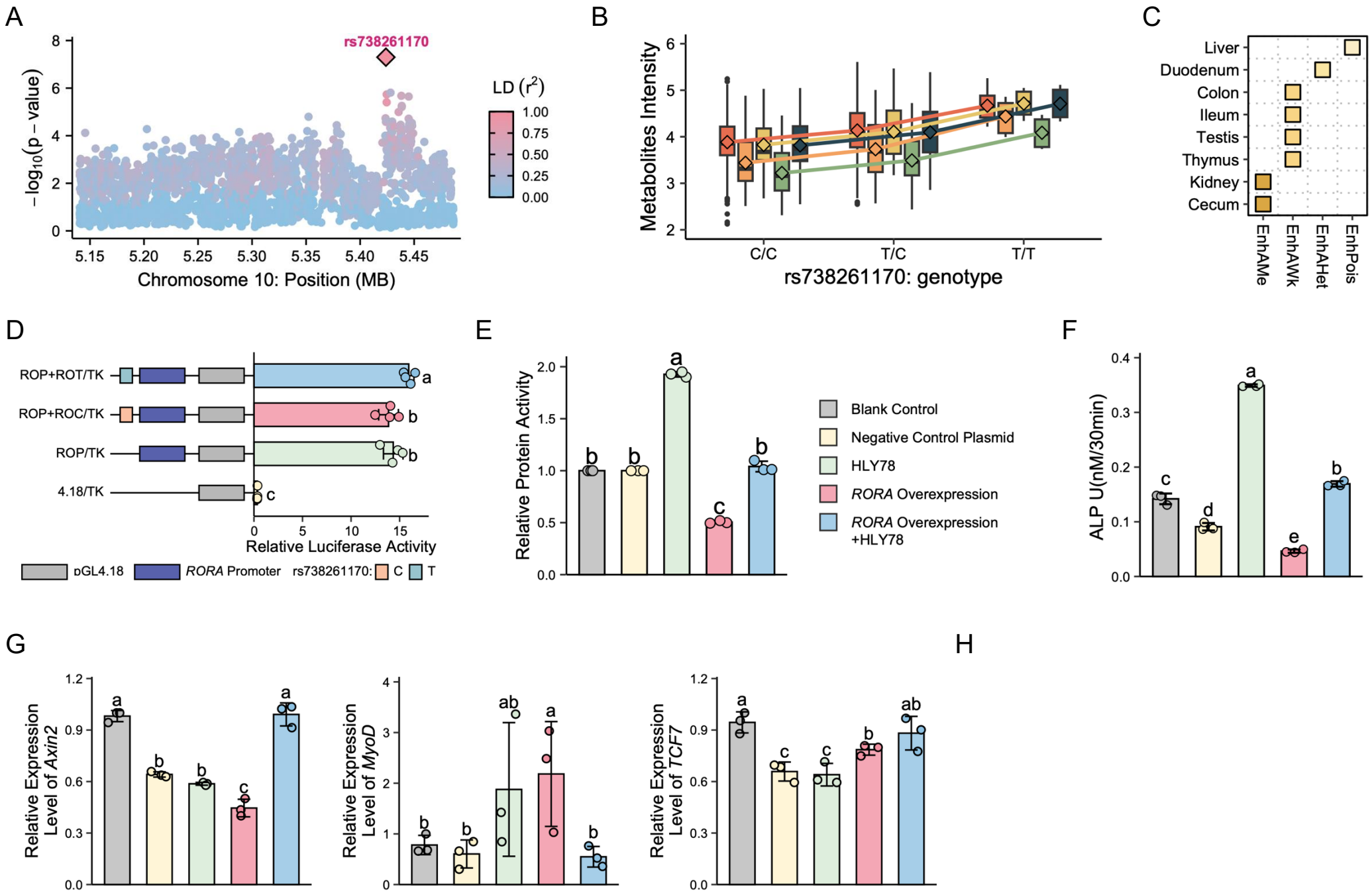

A
B
C
D
E
Bone Strength (N)
Bone Mineral Density (g/cm³)
Bone Ash (g/g)
Days of Age
Group
CA
CT
F
Average Daily Gain (g)
0-28
28-42
G
Tibial Bone Weight (g)
Tibial Bone Diameter (mm)
Tibial Bone Length (mm)
H
Body Weight (Kg)
I
Bone Calcium (%)
Serum Calcium (mmol/L)
J
ALP (King's unit/100ml)
OPG (pmol/L)
K
Bone Phosphorus (%)
Serum Phosphorus (mmol/L)
L
Increase in Node Purity
P
OPG
ALP
VD3
PTH
PGE
CT
OC
Ca
Feature


A
Super Class
Phenylpropanoids and polyketides
Organic oxygen compounds
Benzenoids
Other
Organoheterocyclic compounds
Organic acids and derivatives
Lipids and lipid-like molecules
4.21%
7.50%
8.62%
14.50%
15.83%
16.77%
32.58%
0.00 0.25 0.50 0.75 1.00
Percentage
B
To1
Tp1 (5.48%)
Group
CA
CT
C
Number
CA CT
645 784
1355 1391
930 837
570 499
337 310
654 670
0-20
20-40
40-60
60-80
80-100
>100
CV(%)
D
Significant Metabolites
-log2(p-value)
VIP
E
Super Class
FC
lg(VIP)
CV
F
KEGG Pathway
−log2 (p-value)
Citrate cycle (TCA cycle)
Lipoic acid metabolism
Adrenergic signaling in cardiomyocytes
beta-Alanine metabolism
Butanoate metabolism
Pyrimidine metabolism
Arginine and proline metabolism
Enrichment Score
G
Metabolite Count
Phenylpropanoids and polyketides
Organic oxygen compounds
Benzenoids
Other
Organoheterocyclic compounds
Organic acids and derivatives
Lipids and lipid-like molecules
OPG
BS
TBD
TBL
TBW
P
Ca
BMD
ALP
Set Intersections
H
Isobutyryl-L-Carnitine
Terpyridine
Fluroxene
Intensity
Group
I
Panaxydol
Glucuronic Acid Sulfate
Dihydroergotamine

A
Metabolite Super Class
Lipids and lipid−like molecules
Organic acids and derivatives
Organoheterocyclic compounds
Benzenoids
Phenylpropanoids and polyketides
Organic oxygen compounds
Other
−log10(p−value)
Chromosome
B
Proportion of All QTL
Proportion of mQTL
Intron
Intergenic
Upstream
Downstream
Synonymous
3 prime UTR
5 prime UTR
Proportion(%) of SNP
C
Percentage(%)
Super Calss
Heritability
>0.5
0.4-0.5
0.3-0.4
0.2-0.3
0.1-0.2
0.0-0.1
D
Qui
ReprWk
Repr
TssBiv
ATAC_Is
EnhPois
EnhAHet
EnhAWk
EnhAMe
EnhA
TxFlnkHet
TxFlnkWk
TxFlnK
TssAHet
TssA
Enrichment Factor
E
Lambda (λ)
1.05
0.95
Super Class
F
Number (k)
mQTL
Gene
1156
664
312
192
180
98
117
68
11
6
41
18
17
13
Super Class
G
Transcription Factors
Genes
Metabolites
H
Average number of TFs=23
Density
The Number of TFs
J
Gene Number
115
96
96
76
75
82
82
82
82
82
DEAF1
ARID3A
ARNT
SIX4
PAX2


A
Metabolite Super Class
Lipids and lipid-like molecules
Organic acids and derivatives
Organoheterocyclic compounds
Benzenoids
Phenylpropanoids and polyketides
Organic oxygen compounds
Other
chCADD
phastCons
B
Metabolite-Gene-Developmental Disorders
Metabolite-Gene-Skeletal Disorders
Node
Gene
Developmental Disorders
Skeletal Disorders
C
Gene Ontology Term
Cardiac septum morphogenesis
ATP hydrolysis activity
Response to sterol
Semi-lunar valve development
Atrial septum morphogenesis
Central nervous system neuron differentiation
Cardiac septum development
Pulmonary valve morphogenesis
Heart valve morphogenesis
Cardiac chamber morphogenesis
−log10(p − value)
SMAD binding
Receptor complex
Regulation of membrane potential
Receptor serine/threonine kinase activity
Somatodendritic compartment
Regulation of postsynaptic membrane potential
Synaptic signaling
Glutamate receptor activity
Plasma membrane protein complex
ATP-dependent activity
Onthology
Biological Process
Cellular Component
Molecular Function
D
108 (18.2%)
20 (3.4%)
464 (78.4%)
Chicken
Human
E
Gene Name
KIF1A
FBN1
UCHL1
TGFBR2
SMAD6
PIEZO2
BMPR1B
ATP8A2
Associated Diseases Count
F
Gene Count
G2P: DD
G2P: SD
IPMC
MGI
G
−log100(p − value)
M16
M07
M10
M43
M15-M19
M67
M53
M85
M89
M95-M99
M86
M72
BMPR1B
FRYL
IFIH1
KIF1A
KLHL40
MAF
MAGI2
NOTCH2
PIEZO2
SNX14
TGFBR2
TRAF3IP3
WHRN
WWOX
Trait Class
Other Traits
Leg Disease Related Traits


A
L-Carnitine
Acetyl-CoA
L-Acetylcarnitine
Deacetylation
CAT
TCA cycle
ATP
Blood-brain barrier
ACh
NF-κB
Nrf2/ARE
2-Hydroxyhexadecanoic acid
CPT1/2
2-Hydroxyhexadecanoylcarnitine
β-oxidation
ROS
Caspase-3
Fatty acid
CPT2
Acyl carnitine
Isobutyric acid
CPT1B
ACAT1
Isobutyryl-L-carnitine
Propionyl-CoA
Gluconeogenesis
PPARα/γ
B
L-Carnitine
2-Hydroxyhexadecanoylcarnitine
Isobutyryl-L-carnitine
L-Acetylcarnitine
Intensity
CA
CT
C
TTC7A
rs15305076
rs315053015
−log10(p−value)
PIP(%)
LD (r2)
Chromosome 3: Position (MB)
D
Intensity
A/A
A/G
G/G
E
Percentage(%)
Genotype
G/G
A/G
A/A
F
T/T
T/C
C/C
G
C/C
T/C
T/T
H
TPM of TTC7A
I
GO Terms Significantly enriched in Transcription Factor Gene Sets
Central Nervous System Development
Transcription Factor Binding
Regulation of Cell Population Proliferation
Transcription Regulator Complex
Animal Organ Development
Anatomical Structure Morphogenesis
Animal Organ Morphogenesis
Cell Population Proliferation
Embryo Development
Brain Development
Pattern Specification Process
Head Development
p-value
Similarity
Counts


A
rs738261170
LD (r^2)
1.00
0.75
0.50
0.25
0.00
$-\log_{10}$(p – value)
Chromosome 10: Position (MB)
B
Metabolites Intensity
C/C
T/C
T/T
rs738261170: genotype
C
Liver
Duodenum
Colon
Ileum
Testis
Thymus
Kidney
Cecum
EnhAMe
EnhAWk
EnhAHet
EnhPois
D
ROP+ROT/TK
ROP+ROC/TK
ROP/TK
4.18/TK
Relative Luciferase Activity
pGL4.18
RORA Promoter
rs738261170: C T
E
Relative Protein Activity
Blank Control
Negative Control Plasmid
HLY78
RORA Overexpression
RORA Overexpression +HLY78
F
ALP U(nM/30min)
G
Relative Expression Level of Axin2
Relative Expression Level of MyoD
Relative Expression Level of TCF7
H